

KOMBINI VENDING SYSTEM

DRINK LOGIC + USER FLOW (INTEGRATED VERSION)

1. HIGH-LEVEL PHILOSOPHY

- The vending machine behaves like a **café with a fixed menu**, not a drink builder.
- Users always start from **named, predefined drinks**.
- The **machine logic is modular**, but the **UI never exposes raw building blocks**.
- Customization is **safe, predictable, and limited**.
- The system clearly communicates:
 - what is included
 - what is optional
 - what cannot be changed

2. DRINK CATEGORIES (SYSTEM LEVEL)

2.1 Matcha-Based Drinks

Core drinks:

- Classic Matcha Latte
- Iced Matcha Latte
- Matcha Tonic
- Matcha Lemonade
- Dirty Matcha (Matcha Latte + Espresso Shot)

Rules:

- Matcha quantity is **fixed**
- No “extra matcha”
- No “matcha shot”

2.2 Coffee-Based Drinks

Core drinks:

- Espresso
- Americano
- Classic Latte
- Iced Latte
- Iced Coffee
- Espresso Tonic

Rules:

- Espresso is the base
- Extra espresso shot is allowed **only** for espresso-based milk drinks (e.g. Latte)

2.3 Flavor Variations (Syrup-Based)

Flavor variations are menu extensions, not custom builds.

They are created by:

- taking a core drink
- adding a predefined syrup

Matcha Flavor Variants:

- Strawberry Matcha Latte
- Mango Matcha Latte
- Vanilla Matcha Latte
- Caramel Matcha Latte
- Coconut Matcha Latte
- Lavender Matcha Latte
- Taro Matcha Latte

Coffee Flavor Variants:

- Vanilla Latte

- Caramel Latte
- Shoyu Caramel Latte
- Mango Iced Coffee
- Vanilla Iced Latte
- Caramel Iced Latte

These are **named drinks**, not user-constructed recipes.

2.4 Iced / Layered Drinks (Visual Category)

Purpose:

- visual appeal
- social / Instagram content
- layered presentation

Examples:

- Iced Taro Matcha Latte
- Iced Strawberry Matcha Latte
- Iced Vanilla Matcha Latte
- Iced Caramel Matcha Latte
- Iced Mango Matcha Latte
- Iced Vanilla Latte
- Iced Caramel Latte
- Iced Shoyu Caramel Latte

System rule:

- These are **presentation variants** of existing drinks
- Logic is identical to their non-layered counterparts
- Difference is **visual / preparation order only**

3. MODULAR MACHINE LOGIC (INTERNAL ONLY)

The machine internally operates on a modular system, but this is **never exposed** directly to the user.

3.1 Internal Components

BASE

- Espresso
- Matcha

LIQUID

- Water
- Milk (Cow / Oat, powdered)

STYLE

- Hot
- Iced

FLAVOR (Syrups / Powders)

- Vanilla
- Caramel
- Strawberry
- Mango
- Taro
- Shoyu Caramel

3.2 Internal Formula

One drink internally equals:

[Base] + [Liquid] + [Style] + [Flavor]
Example:

Iced Taro Matcha Latte

- Matcha powder
- Taro syrup / powder
- Milk powder
- Water

→ Ice

Important:

- This modularity is for the machine and software
- The user always sees a named drink

4. DAY-ONE MENU (LOCKED & CLEAN)

To ensure:

- speed
- stability
- low error rate
- easier user understanding

4.1 Matcha (5)

- Classic Matcha Latte
- Iced Matcha Latte
- Matcha Tonic
- Matcha Lemonade
- Dirty Matcha

4.2 Coffee (5)

- Espresso
- Americano
- Classic Latte
- Iced Latte
- Espresso Tonic

4.3 Flavor Strategy (Day One)

- Flavor syrups are modifiers, not separate menu items
- Rotating flavors (availability-based):
 - Vanilla

- Caramel
- Strawberry
- Mango
- Taro
- Shoyu Caramel

This keeps:

- menu simple
- machine logic flexible
- seasonal rotation easy

5. CUSTOMIZATION RULES

5.1 Sweetness (Universal)

Available for all drinks:

- Less
- Normal (default)
- Extra

No cost.

5.2 Syrups (Coffee + Matcha)

Case A: No Syrup in Base Drink

(e.g. Classic Matcha Latte, Cafe Latte, Matcha Tonic)

UI shows:

- “Add syrup”
- User selects **one flavor**
- Optional: add extra pumps

System:

- Syrup is marked as **Extra**

Case B: Syrup Already in Base Drink

(e.g. Strawberry Matcha Latte)

UI shows:

- “Base syrup: Strawberry (1 pump)”

Allowed actions:

- “Add more” (same syrup)
- Optional secondary action: “Add another syrup”

System:

- Base syrup and extra syrups are stored separately
- No duplicate confusion
- No accidental over-sweetening

5.3 Extra Espresso Shot

- Available only for:
 - Cafe Latte
 - Vanilla Latte
 - Caramel Latte
- Always paid
- Never shown for matcha drinks

6. WHAT IS EXPLICITLY NOT ALLOWED

- Building drinks from scratch
- Extra matcha
- Removing base ingredients
- Changing ingredient order
- Mixing incompatible bases
- Guessing whether a syrup is included

7. WHY THIS SYSTEM WORKS

- Matches real café behavior
- Addresses manufacturer's concerns
- Prevents bad taste combinations
- Keeps flow fast (2–3 decisions)
- Scales easily with new flavors
- Is simple to implement in software
- Protects the brand

USER FLOW

STATE 0: IDLE / ATTRACT MODE

- Fullscreen animated visuals
- Brand characters
- No buttons
- User touches screen



Transition to State 1

STATE 1: MENU — DRINK SELECTION

- Display predefined drinks
- Categories:
 - Matcha
 - Coffee
 - Hybrid (Dirty)
- Each drink shows:
 - Name
 - Price
- No ingredient lists

User taps a drink



Transition to State 2

STATE 2: PRODUCT HERO

- Large product visual
- Drink name
- Price
- Short vibe description (1 line max)

Actions:

- **Add to cart**
- **Customize**

➡ If Customize → State 3

➡ If Add to cart → State 4

STATE 3: CUSTOMIZATION (OPTIONAL)

Visible sections:

1. **Sweetness** (always visible)
2. **Syrups** (dynamic logic based on recipe)
3. **Extra espresso shot** (only for espresso-based drinks)

Rules:

- Drink visual always visible
- Price updates in real time
- All included elements clearly labeled (“Included”)

User confirms customization

➡ Transition to State 4

STATE 4: CART

- Miniature cup thumbnails
- Full breakdown:
 - Drink name
 - Sweetness level
 - Base syrup (if any)
 - Extra syrup pumps (if any)
 - Extra espresso (if any)
- Final price

Action:

- Pay

➡ Transition to State 5

STATE 5: PAYMENT

- Card payment
- Simple, minimal UI
- Retry on failure
- Timeout returns to cart

➡ On success → State 6

STATE 6: PREPARATION

- Drink visually drops from shelf
- Animated preparation sequence
- No user interaction

➡ Transition to State 7

STATE 7: REWARD EVALUATION (BACKGROUND)

- Loyalty points calculation
- Season pass / reward logic

If reward unlocked:

➡ State 8

If not:

➡ State 9

STATE 8: BLINDBOX / REWARD (OPTIONAL)

- Mystery pouch animation
- Reward reveal
- If user not logged in:
 - Temporary save
 - CTA to create account

➡ Transition to State 9

STATE 9: COMPLETE / PICKUP

- Message: “Your drink is ready”
- Visual pickup indicator
- Optional QR for account creation

After timeout:

➡ Return to Idle Mode

7. KEY UX RULE

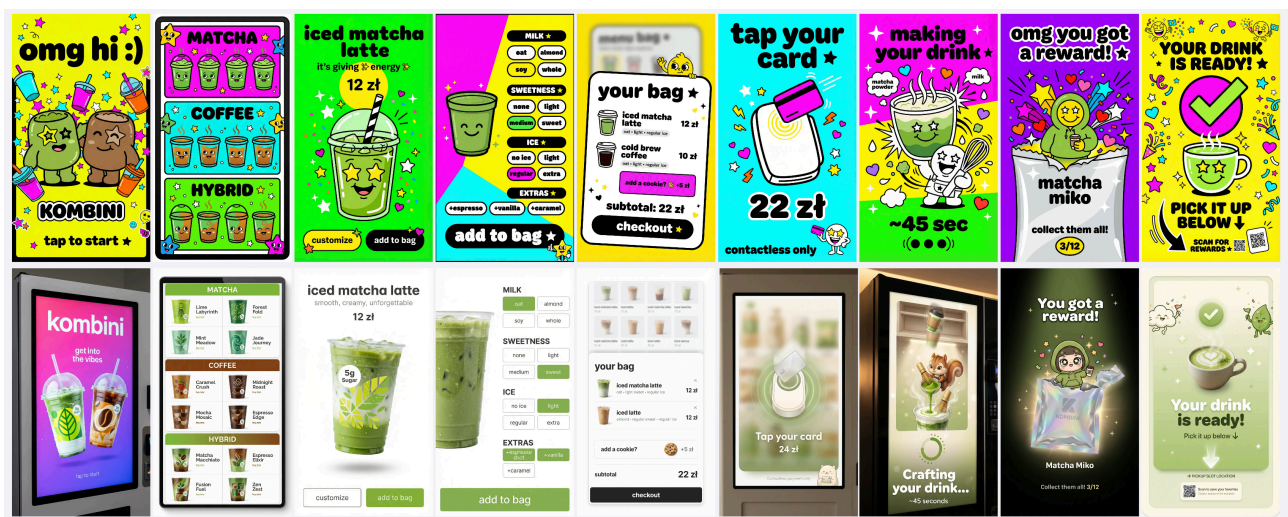
The user should never have to guess what is inside the drink.

The machine must always clearly communicate:

- what is included
- what is extra
- what can be changed

This guarantees:

- predictable taste
- faster ordering
- higher satisfaction
- fewer mistakes



These are two sample visuals showing how the flow could look in different graphic styles, as described in the document above. You can immediately assess whether something as visually complex as the most colorful example can be implemented, or whether it needs to be much simpler.